Interlude: Memory API

Лекция №8

Types of Memory

```
void func() {
    int x; // declares an integer on the stack
    ...
}
```

```
void func() {
    int *x = (int *) malloc(sizeof(int));
    ...
}
```

The malloc() Call

```
#include <stdlib.h>
...
void *malloc(size_t size);
```

```
double *d = (double *) malloc(sizeof(double));
```

```
int *x = malloc(10 * sizeof(int));
printf("%d\n", sizeof(x));
```

```
int x[10];
printf("%d\n", sizeof(x));
```

The free() Call

```
int *x = malloc(10 * sizeof(int));
...
free(x);
```

Common Errors

Forgetting To Allocate Memory

```
char *src = "hello";
char *dst;              // oops! unallocated
strcpy(dst, src); // segfault and die
```

```
char *src = "hello";
char *dst = (char *) malloc(strlen(src) + 1);
strcpy(dst, src); // work properly
```

Common Errors

Not Allocating Enough Memory

```
char *src = "hello";
char *dst = (char *) malloc(strlen(src)); // too small!
strcpy(dst, src); // work properly
```